COMPILER DESIGN

ASSIGNMENT

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Course Name	:	Compiler Design
Slot	:	C
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Short Answers test

Q1: for the grammar

$E \rightarrow E + I \mid T$ remove left recursion

The general transformation is

$$A \rightarrow A\alpha \mid \beta$$

$$A' \rightarrow \alpha A' \mid \epsilon$$

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

Q2: find first (E) for:-

$$E \rightarrow TE', \quad E' \rightarrow +TE' \mid \epsilon, \quad T \rightarrow id$$

given :-

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

first (E) = first (T)

$$T \rightarrow id$$

first (E) = $\{ id \}$

Q3: for the grammar $E \rightarrow T + E \mid T$

ambiguous or not

$$E \rightarrow T + E \mid T$$

$$T \rightarrow T * T$$

$$E \rightarrow T + E \rightarrow T + T + E \rightarrow T + T + T$$

The grammar is unambiguous.

4) Identify whether string aba is valid for:-

$S \rightarrow aSa \mid b$

$S \Rightarrow aSa$

$\Rightarrow aba$

'aba' is valid.

5) Over whether the string aba generated by

$S \rightarrow aba \rightarrow b$

$S \Rightarrow aA$

$A \Rightarrow b$

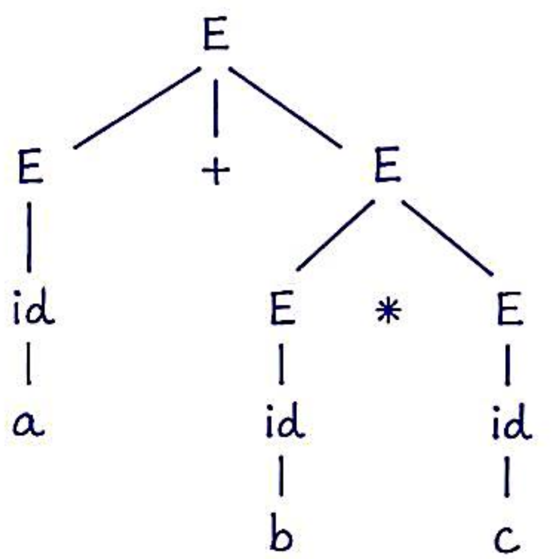
$S \Rightarrow aA$

$\Rightarrow ab$

'ab' is generated.

6) Construct whether the string ab is parse tree for a+b* (using $E \rightarrow E+E \mid E*E \mid id$

$a + (b*c)$ parse tree



$a+(b*c)$

7. Apply left factoring for:-

$$S \to aB \mid aB \mid b$$

The common prefix is a.

$$S \to aA \mid b$$

$$A \to B \mid B$$

$$S \to aA \mid b$$

$$S \to A \mid b$$

8. find follow(A) for:-

$$S \to AB, B \to b$$

$$S \to AB$$

$$B \to b$$

$$\text{follow}(A) = \text{first}(B)$$

$$\text{first}(B) = \{b\}$$

$$\text{follow}(A) = \{b\}$$

9. given the grammar

$$S \to aSb \mid \varepsilon$$

The string is aabbbb is valid using derivation

$$S \Rightarrow aSb \mid \varepsilon$$

$$S \Rightarrow aSb$$

$$\Rightarrow aSbb$$

$$\Rightarrow aaSbbb$$

$$\Rightarrow aaaSbbb$$

$$\Rightarrow aaaaSbbb$$

'aabbbb' is valid

10, Convert the following ambiguous grammar is into an unambiguous grammar:-

$E \to EE \mid E * E \mid id$

$E \to E + T \mid T$

$T \to T * F \mid F$

$\boxed{F \to id}$

This ensures that multiplication has higher precedence than addition.

(F) first = { id }

{ F } = (F) first = { id }

{ E } = (E) follow = { $ }

$ = End of input marker